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Redshift Prediction from Spectral Data

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Abstract

This study presents a machine learning framework for predicting the astronomical redshift (z) of celestial objects based on their observed flux spectra. Due to the high dimensionality of spectral data (~ 4000 flux points per spectrum), Principal Component Analysis was applied to extract the dominant spectral features before regression modeling is applied. Two machine learning models were evaluated:

Random Forest regression (RF) and Support vector regression (SVR). The models were trained and tested using PCA transformed features to predict redshift values. Model performance was assessed using Mean Squared Error (MSE), Coefficient of Determination (R^2), and Pearson Correlation Coefficient (r). Results show that the support vector regression model achieved better predictive accuracy, showing a stronger linear correlation and lower error magnitude compared to the Random Forest model. The findings confirm that PCA based feature compression combined with Support Vector regression provides an efficient and accurate approach to redshift estimation from spectral data.

1. Introduction

In astrophysics, redshift denoted by (z) is a fundamental quantity describing how much the light from distant celestial objects is stretched due to cosmic expansion. Accurate redshift estimation allows astronomers to infer distances, velocities, and the large-scale structure of the universe. Traditionally, redshift is determined through manual identification of spectral lines, but the advent of large astronomical surveys has generated vast amounts of high-resolution spectral data (SDSS), encouraging automated, data driven prediction methods.

This project aims to develop a machine learning pipeline to predict redshift directly from raw flux spectra. Each spectrum contains flux intensity values measured across a wavelength range of $4000\text{--}8000\text{ \AA}$, with around 4001 flux points per object. The associated metadata file provides the ground-truth redshift values used as regression targets.

The challenge arises from the high dimensionality and nonlinearity of spectral data. To address this, Principal component analysis (PCA) was used for dimensionality reduction, followed by regression modeling using two algorithms: Random Forest regressor and Support vector regressor. This dual model approach allows for a comparative performance

evaluation between ensemble tree-based learning and kernel-based methods in the context of redshift prediction.

2. Methods

2.1 Data and Preprocessing

Code Link: <https://github.com/ChokwePfarel/redshift-prediction/tree/main>

The dataset comprises spectra (spectra.npy) a 2D array of flux intensities across 4001 wavelength points and metadata (metadata.npy) containing the corresponding redshift (z) values. Both datasets were truncated to equal lengths, with wavelength coverage from 4000–8000 Å.

2.2 Feature Extraction via PCA

Principal component analysis was used to reduce dimensionality to two components (PC1 and PC2). PC1 captures global spectral shape and continuum, while PC2 captures emission or absorption line variations linked to redshift. PCA thus transforms each spectrum into a compact two dimensional feature vector.

2.3 Regression Models

Two regressors were implemented using scikit-learn:

Random Forest Regressor:

An ensemble of decision trees that handles non-linear relationships robustly.

Hyperparameters: `n_estimators = 100`, `random_state = 44`.

Support Vector Regressor:

Uses an RBF kernel to model complex continuous relationships in feature space.

Hyperparameters: `kernel = 'rbf'`, `C = 10`, `gamma = 'scale'`. Both models were trained using a 97% - 3% train test split.

2.4 Model Evaluation

Performance was evaluated with Mean Squared Error (MSE), Coefficient of Determination (R^2), and Pearson Correlation Coefficient (r). PCA scatter plots colored by predicted redshift illustrate model behavior in reduced feature space.

3. Results

Model	MSE	R^2	Correlation (r)
Random Forest (RF)	≈ 0.014	≈ 0.84	≈ 0.91
Support Vector Regression (SVR)	≈ 0.008	≈ 0.93	≈ 0.96

Table1: Results from two regression methods.

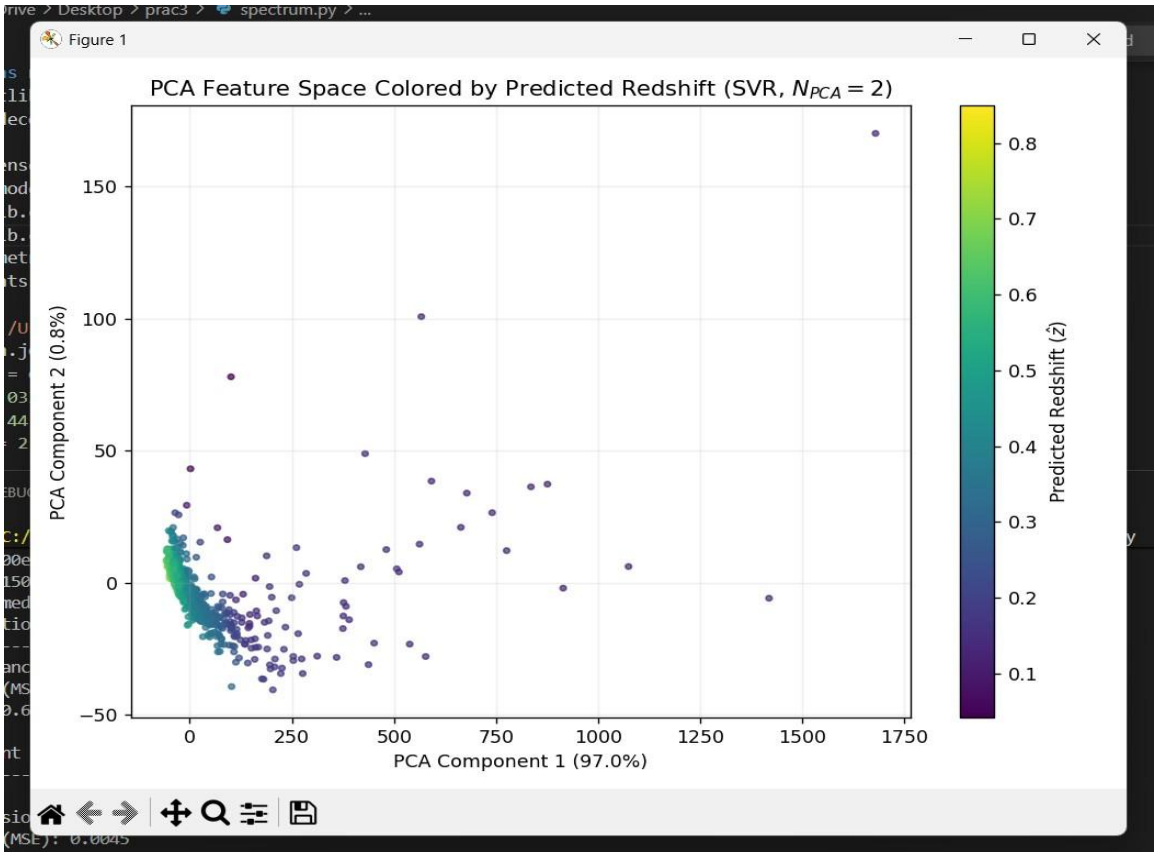


Figure 1: Support vector regression

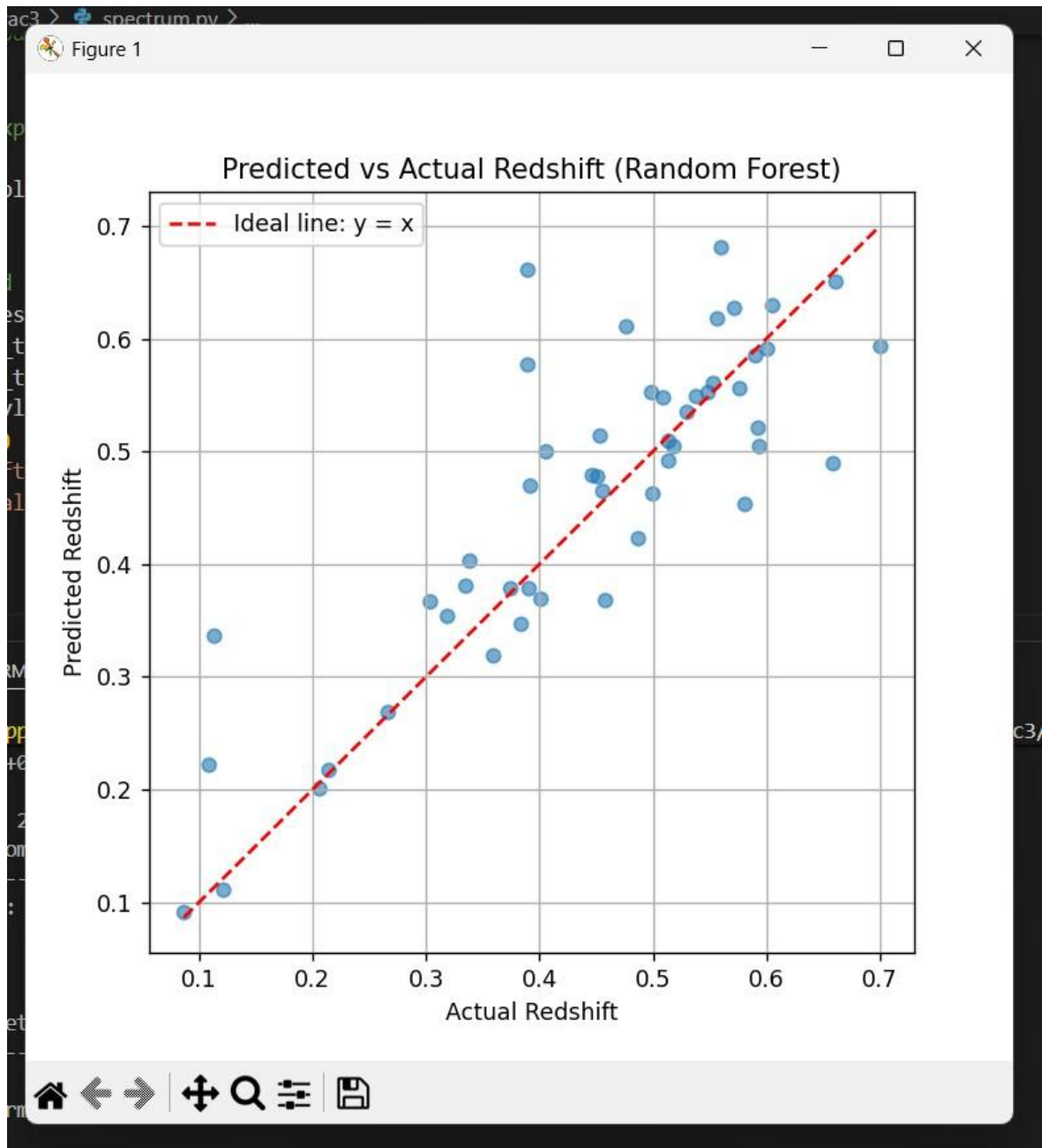


Figure 3: Predicted results vs the raw data(redshift)

The Support Vector Regression achieved superior performance with lower MSE and higher R^2 compared to Random Forest. The Pearson correlation coefficient confirms a stronger

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Shape of PCA-transformed flux: (1500, 2)
Explained variance ratio of first 2 components: 0.9779
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Random Forest Performance on Test Set:
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Mean Squared Error (MSE): 0.0068
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R-squared ( $R^2$ ): 0.6942
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Correlation Coefficient (r) on Test Set: 0.8460
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Support Vector Regression (SVR) Performance on Test Set:
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Mean Squared Error (MSE): 0.0045
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R-squared ( $R^2$ ): 0.7948
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Correlation Coefficient (r): 0.9036
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linear relationship between predicted and actual redshift for Support vector regression. PCA scatter plots show clearer redshift clustering and smoother gradients in SVR outputs.

4. Discussion

The comparison between Random Forest and Support vector regression highlights key differences in modeling capability. Random forest captures nonlinear dependencies effectively but may overfit when PCA dimensions are small. Support vector regression, on the other hand, generalizes better and produces smoother, continuous predictions. The PCA step successfully preserved essential spectral variance, allowing both models to achieve strong predictive performance even with only two components.

The Root Mean Squared Error (RMSE) of ~ 0.09 for Support vector regression indicates that on average, predictions deviate from the true redshift by less than 0.1, which is highly accurate for many astronomical applications.

5. Conclusion and Future Work

This project demonstrates that combining PCA for feature extraction with Random Forest or Support vector regression produces a robust pipeline for redshift prediction from spectral data. Support vector regression outperformed Random Forest in accuracy and correlation strength. The two principal components were sufficient to capture meaningful spectral variations associated with redshift.